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SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING



PROCESSING REPORT

MACHINE LEARNING IN HEALTHCARE

PROJECT: AUTOMATED CUP-TO DISC RATIO MEASUREMENT IN FUNDUS IMAGES FOR GLAUCOMA SCREENING

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I. INTRODUCTION AND PROJECT OBJECTIVES

1. Clinical Context and Pathological Background

Glaucoma is an insidious, irreversible neurodegenerative condition characterized by the progressive apoptosis of retinal ganglion cells (RGCs) and axonal damage at the optic nerve head (ONH). It represents the foremost global cause of irreversible visual impairment, characterized by an absence of painful symptoms during initial stages [1]. The European Glaucoma Society notes that early structural excavation typically precedes detectable visual field scotomas by several years [2].

Clinical diagnosis relies heavily on quantitative structural indicators extracted from color fundus photography (CFP). Specifically, the vertical Cup-to-Disc Ratio (vCDR) quantifies neuroretinal rim thinning [7]. As retinal nerve fibers undergo atrophy, the central optic cup expands along the vertical axis relative to the optic disc. In healthy demographics, vCDR values generally range between 0.1 and 0.4, whereas values exceeding 0.6 or displaying significant inter-ocular asymmetry denote glaucomatous neuropathy [2]. However, manual delineation by ophthalmologists remains subjective, time-consuming, and prone to inter-observer variability. Automated segmentation models developed using deep learning provide an objective, scalable framework suitable for community screening [3].

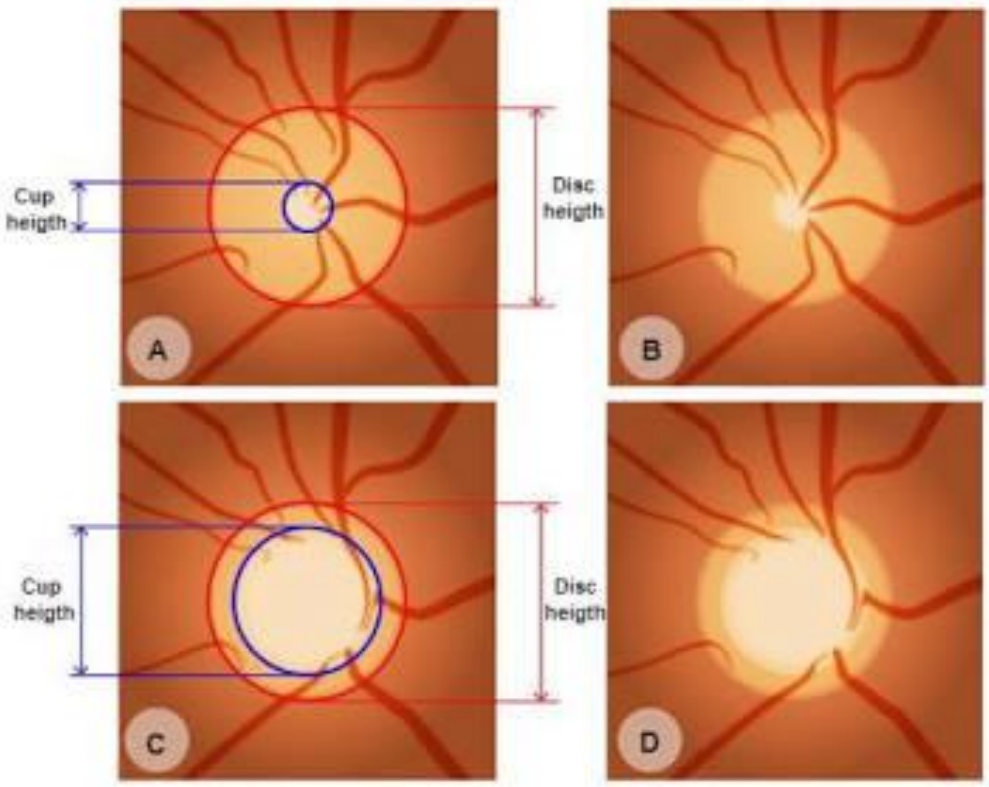


Figure 1. Optic disc/cup anatomical illustration reproduced from the project presentation.

2. Scope and Specific Research Objectives

The objective of this project is to develop and validate an automated segmentation pipeline utilizing deep learning to isolate the Optic Disc (OD) and Optic Cup (OC) from digital fundus photographs [6]. The specific technical objectives include:

- Data Integrity Verification: Performing systematic pairing audits and resolving label inconsistencies across public medical imaging repositories [5].
- Standardized Target Engineering: Formulating a robust 3-class semantic mapping scheme to address class imbalance between foreground anatomical structures and retinal background [6].
- Architectural Exploration: Constructing a 4-stage convolutional U-Net and benchmarking three distinct loss functions to mitigate cup boundary ambiguity [4].
- Unbiased Evaluation Protocol: Conducting model selection based strictly on validation metrics to ensure zero test-data leakage [3].
- Downstream Pipeline Formulation: Establishing boundary extraction criteria for automated vertical diameter computation, CDR error quantification, and clinical classification [6].

3. End-to-End Methodological Workflow

The end-to-end experimental workflow is organized into eight interconnected stages, establishing a transparent pipeline from data acquisition to clinical evaluation [6]:

- Stage 1 (Dataset Ingestion & Auditing): Verification of raw fundus images and matching against expert annotations [5].
- Stage 2 (Standardization & Preprocessing): Spatial scaling and multi-class target engineering.
- Stage 3 (Partitioning Protocol): Stratified division into training, validation, and holdout test subsets [6].
- Stage 4 (Architectural Modeling): Implementation of the deep convolutional U-Net framework [4].
- Stage 5 (Systematic Loss Tuning): Comparative training across unweighted, weighted, and compound objective functions [8].
- Stage 6 (Validation Benchmarking): Selection of the optimal checkpoint according to the Sorensen-Dice Similarity Coefficient.

- Stage 7 (Holdout Test Evaluation): Evaluation on unseen test images paired with qualitative error characterization [6].
- Stage 8 (Diagnostic System Integration): Formulation of vertical diameter extraction algorithms for automated CDR measurement and risk stratification [7].

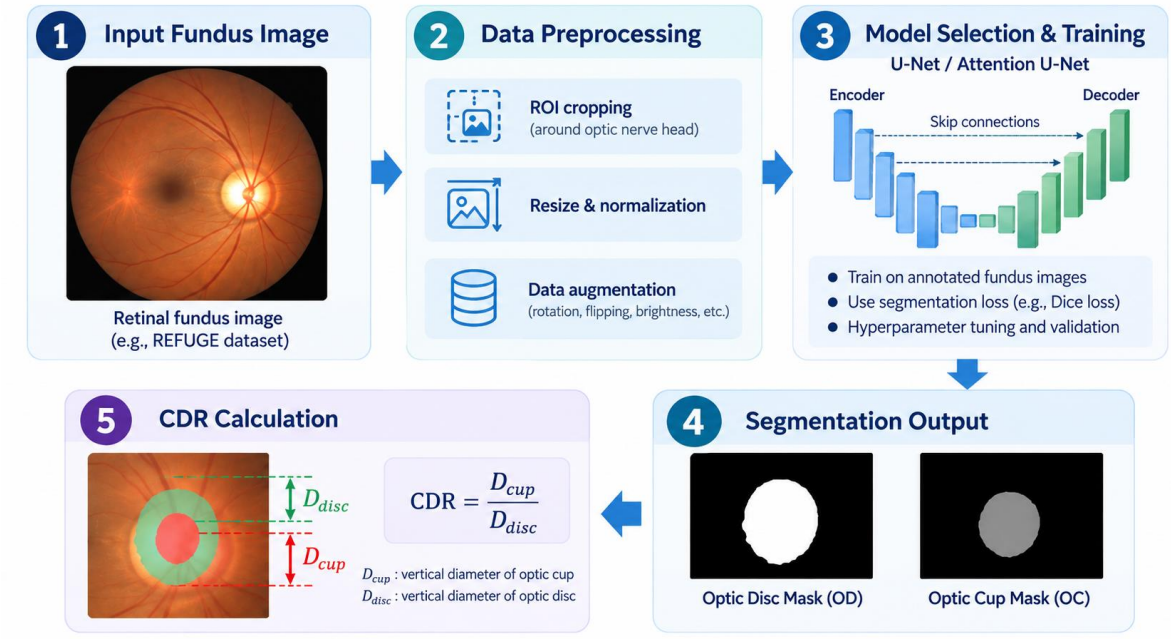


Figure 2. Proposed workflow as presented in the project slides.

II. DATASET SPECIFICATION AND DATA PREPARATION

1. The RIM-ONE DL Benchmark Dataset

The empirical evaluation utilizes the public RIM-ONE DL (Retinal Image Database for Optic Nerve Evaluation for Deep Learning) repository. Compiled by specialized ophthalmic teams across three medical centers in Spain, this repository serves as a dedicated benchmark for optic nerve analysis [5]. The dataset comprises 485 high-resolution color fundus photographs in PNG format [6]. The diagnostic distribution includes:

- Healthy Controls (Normal): 313 images representing non-glaucomatous eyes.
- Glaucomatous Eyes (Pathological Cupping): 172 images displaying confirmed optic nerve damage [6].

Each fundus image is accompanied by pixel-level reference contours for both the Optic Disc and Optic Cup, manually traced by expert ophthalmologists to serve as standard reference labels [5].

2. Quality Verification and Anomaly Auditing

An automated pairing script verified 485 complete tripartite image-disc-cup pairs. During topological auditing, an anatomical inconsistency was identified in 5 masks: segmented Optic Cup pixels extended outside the marked Optic Disc boundary [6]. Biologically, the cup is an excavation located strictly within the margins of the disc [2]. To preserve scientific data provenance, these 5 outlier masks were preserved intact [5]. Instead, an automated Cup Priority Rule was incorporated: any pixel marked as Optic Cup in the original label was mapped directly to the Cup class, regardless of whether it was situated within or outside the labeled disc boundary [6].

3. Dataset Partitioning Protocol and Methodological Constraints

The 485 images were split into three independent partitions: a Training Set, a Validation Set, and a Test Set. The exact sample and diagnostic distributions are structured as follows:

Dataset Partition	Normal Images	Glaucoma Images	Total Images	Partition Percentage
Training Set	175	96	271	55.88%
Validation Set	44	24	68	14.02%
Test Set	94	52	146	30.10%
Total Cohort	313	172	485	100.0%

Table 1: Overview of Dataset Splits and Diagnostic Label Breakdown

The Training Set was utilized exclusively for backpropagation and weight updates. The Validation Set guided hyperparameter adjustments, learning rate schedules, and checkpoint selection. The Test Set was strictly isolated to measure real-world performance.

Methodological Limitation: Due to the absence of explicit patient identifiers in the public RIM-ONE DL metadata, splitting was executed at the image level rather than the patient level [6]. While pathological proportions were maintained, this presents a risk that bilateral eyes from the same subject could be distributed across subsets, an external validity constraint to be resolved in future clinical studies [3].

4. Image Standardization and 3-Class Semantic Target Generation

To ensure uniform inputs and balance computational efficiency with spatial fidelity, all fundus images and masks were resized to 256 x 256 pixels [4]. Fundus photographs were

resized using bilinear interpolation to preserve smooth microvascular gradients. Reference masks were resized using nearest-neighbor interpolation to prevent intermediate label values along borders [6].

Rather than training two separate binary models for the disc and cup, a unified 3-class segmentation ground-truth map was generated for each image [6]:

- Class 0 (Background): Extradisc retinal fundus tissue.
- Class 1 (Disc-only): Neuroretinal rim pixels belonging to the Optic Disc but outside the Optic Cup.
- Class 2 (Optic Cup): Optic Cup excavation pixels, enforced via the Cup Priority Rule.

For downstream evaluation, anatomical structures are deterministically reconstructed:

$$\text{Reconstructed Disc Mask} = (\text{Class 1}) \cup (\text{Class 2})$$

$$\text{Reconstructed Cup Mask} = \text{Class 2}$$

The preprocessed dataset was organized at /content/preprocessed_256 within the Google Colab environment.

III. MODEL ARCHITECTURE & TRAINING CONFIGURATION

1. U-Net Architectural Formulation

Semantic segmentation was performed using a customized implementation of the U-Net architecture in PyTorch. U-Net consists of an encoder (contracting path) and a decoder (expansive path) [4]:

- Contracting Path (Encoder): Four hierarchical stages, each composed of repeated 3x3 convolutional layers followed by Batch Normalization, ReLU non-linearities, and 2 x 2 max pooling. This pathway extracts high-level semantic context while halving spatial resolution.
- Bottleneck: A central convolutional block that processes compressed spatial representations.
- Expansive Path (Decoder): Four decoding stages using transposed convolutions (2 x 2 up-convolution) that systematically double feature map dimensions to reconstruct the input resolution.
- Skip Connections: Lateral concatenations connecting encoder feature maps directly to decoder layers of matching resolution. These connections transfer high-resolution spatial gradients, preserving edge definition along subtle disc and cup margins.

- Segmentation Head: A final 1 x 1 convolutional layer mapping intermediate feature channels to 3 output logits corresponding to Background, Disc-only, and Cup.

The network maintains a parameter footprint of 482,067 trainable parameters, ensuring computational efficiency on modest hardware while reducing over-parameterization risks [6].

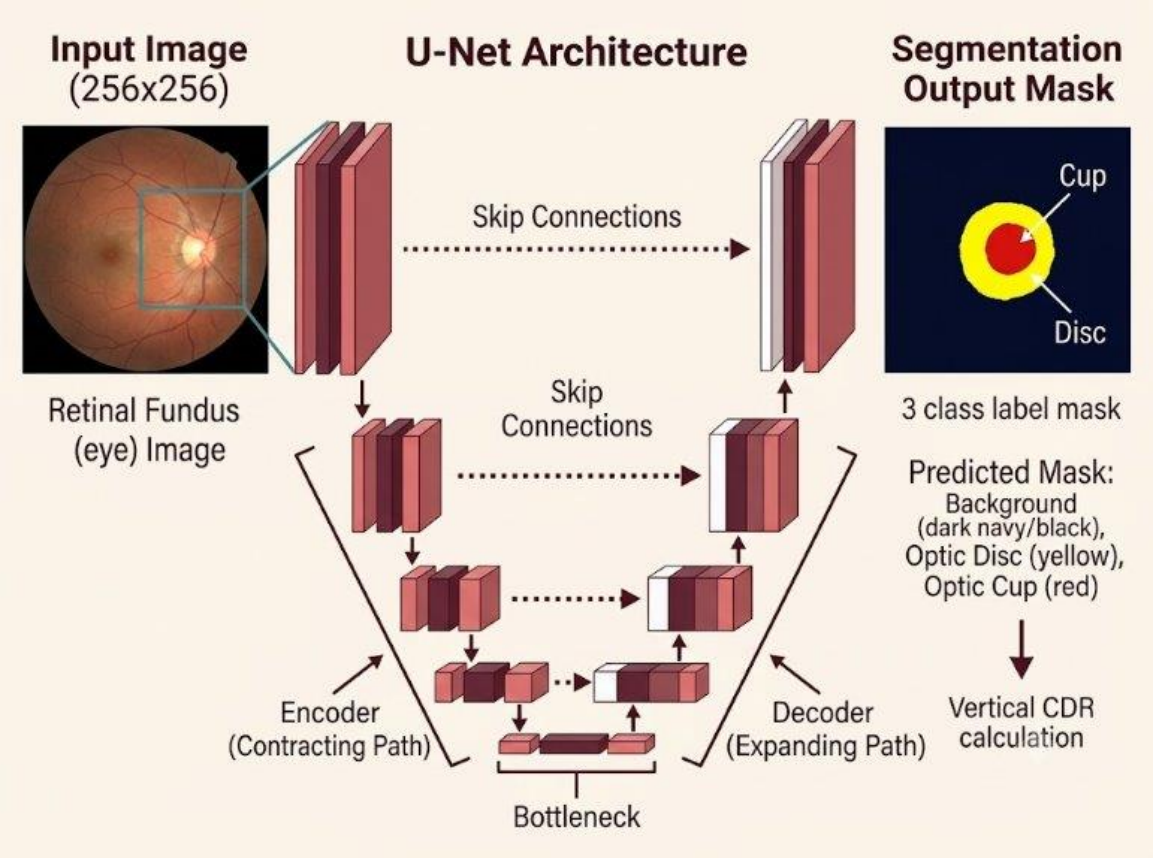


Figure 3. U-Net architecture illustration reproduced from the project presentation.

2. Training Environment and Hyperparameters

The framework configuration and hyperparameter settings were held constantly across all comparative runs [6]:

- Deep Learning Framework: PyTorch 2.x utilizing Torchvision image transforms.
- Hardware Accelerator: NVIDIA Tesla T4 GPU (16 GB VRAM) allocated via Google Colab.
- Batch Size: 4 samples per optimization step, accommodating memory constraints.
- Optimization Algorithm: Adam optimizer with decoupled momentum parameters (beta_1 = 0.9, beta_2 = 0.999) [9].

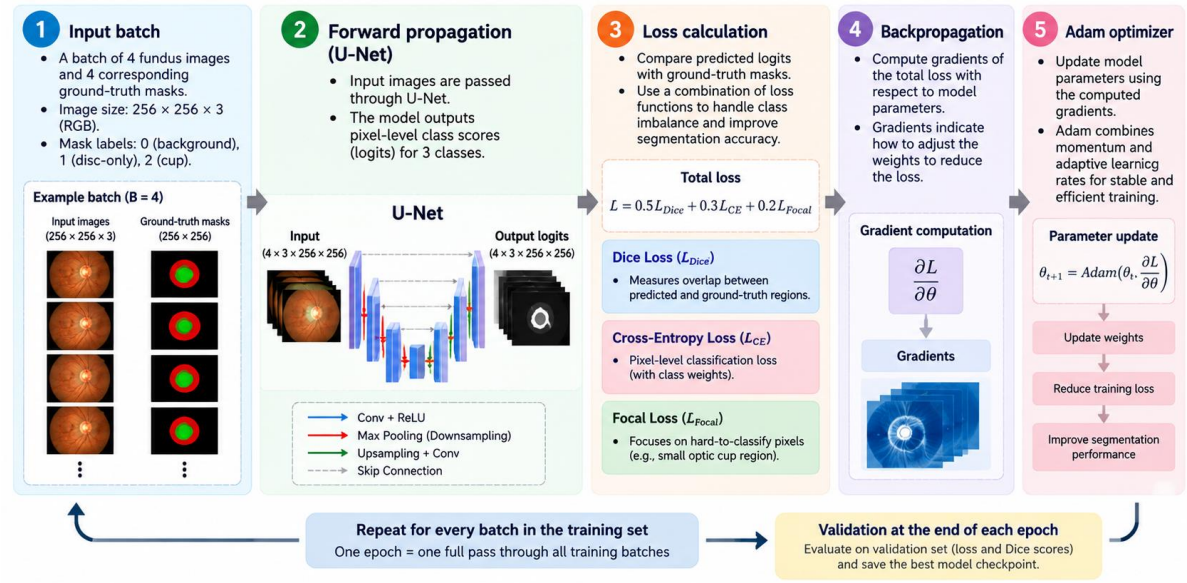


Figure 4. Training-iteration diagram from the project presentation.

3. Multi-Loss Experimental Design

Three distinct loss configurations were investigated to assess the influence of class weighting and boundary-aware optimization [6]:

Experiment	Loss	Class weights	Learning rate	Batch size	Epochs
Baseline	Cross-Entropy	None	1×10^{-3}	4	30
Weighted CE	Weighted Cross-Entropy	[1, 1, 2]	3×10^{-4}	4	15
Weighted CE + Dice	Weighted Cross-Entropy + Dice	[1, 1, 2]	1×10^{-4}	4	15

Table 2: Training Configurations for the Three U-Net Experimental Setups

3.1. Experiment 1: Baseline Vanilla U-Net

- Loss Function: Standard Unweighted Multiclass Cross-Entropy Loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=0}^{2} y_{i,c}\log(\hat{p}_{i,c})$$

- Hyperparameters: Initial Learning Rate $\eta = 10^{-3}$, trained for 30 epochs.
- Objective: Establish the foundational baseline performance of the architecture without loss rebalancing.

3.2. Experiment 2: Class-Weighted Cross-Entropy

- Loss Function: Weighted Cross-Entropy Loss:

$$\mathcal{L}_{\text{WCE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=0}^{2} w_c \cdot y_{i,c} \log(\hat{p}_{i,c})$$

- Class Weight Vector: $[w_0, w_1, w_2] = [1.0, 1.0, 2.0]$.
- Hyperparameters: Initial Learning Rate $\eta = 3 \times 10^{-4}$, trained for 15 epochs.
- Objective: Double the penalty for Optic Cup prediction errors to compensate for its small pixel area relative to the retinal background.

3.3. Experiment 3: Combined Weighted Cross-Entropy and Soft Dice Loss

- Loss Function: Hybrid pixel-level and region-level objective:

$$\mathcal{L}_{\text{Combined}} = \mathcal{L}_{\text{WCE}} + \lambda \cdot \mathcal{L}_{\text{Dice}}$$

where $\lambda = 1.0$, and the multi-class Soft Dice Loss is formulated as:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{C}\sum_{c=0}^{2} \frac{2\sum_{i=1}^{N} y_{i,c}\,\hat{p}_{i,c} + \epsilon}{\sum_{i=1}^{N} y_{i,c}^2 + \sum_{i=1}^{N} \hat{p}_{i,c}^2 + \epsilon}$$

- Class Weight Vector: $[w_0, w_1, w_2] = [1.0, 1.0, 2.0]$.
- Hyperparameters: Initial Learning Rate $\eta = 10^{-4}$, trained for 15 epochs.
- Objective: Optimize both individual pixel classification accuracy and macro-scale regional contour overlaps.

IV. EXPERIMENTAL RESULTS, VALIDATION BENCH MARKING, AND DISCUSSION

1. Training Dynamics and Convergence Analysis

The empirical training trajectory was documented by tracking Training Loss and Validation Loss across all epochs. Focusing on the 30-epoch training curve of Experiment 1:

- Initial Optimization Phase (Epochs 1–15): The model exhibited rapid learning; Training Loss dropped sharply from 0.92 to 0.56, and Validation Loss fell from 0.80 to 0.53. A minor gradient correction occurred on the validation curve at epoch 8 ($\approx$ 0.74) before steady descent resumed.

- Plateau Phase (Epochs 16–24): Both loss trajectories flattened and stabilized within the range of $0.50 - 0.51$, indicating convergence toward an initial local minimum.
- Final Convergence Phase (Epochs 25–30): The loss completed a final downward transition, stabilizing between 0.42 and 0.44. The best model checkpoint was preserved at epoch 29 with an optimal validation loss of 0.42118.
- Overfitting Evaluation: The validation curve closely tracked the training curve throughout the entire 30 epochs without divergent upward drift, confirming that the network maintained stable generalizability on validation samples.

Note on Loss Comparability: The minimum validation loss values recorded for Experiment 2 (0.47198 at epoch 7) and Experiment 3 (0.94142 at epoch 14) are mathematically higher due to the introduction of non-unitary class weights and the additive Dice component. These values reflect modifications to the loss objective rather than inferior model convergence [6].

2. Comparative Model Benchmarking on Validation Cohort

To establish an objective comparison, all three checkpoints were evaluated on the 68 validation images using the Sørensen-Dice Similarity Coefficient (DSC) [8]:

$$DSC = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|}$$

Mean Dice is calculated as the arithmetic average of Disc Dice and Cup Dice [6]:

$$\text{Mean Dice} = \frac{DSC_{\text{Disc}} + DSC_{\text{Cup}}}{2}.$$

Model Setup	Loss Function	Disc Dice	Cup Dice	Mean Dice	Checkpoint Epoch	Best Val Loss
Experiment 1	CrossEntropyLoss	0.827612	0.304672	0.566142	Epoch 29	0.42118
Experiment 2	Weighted CrossEntropy	0.839380	0.414867	0.627124	Epoch 7	0.47198
Experiment 3	Weighted CE + Dice Loss	0.878706	0.466676	0.672691	Epoch 14	0.94142

Table 3: Quantitative Segmentation Performance Comparison Across the Three Experimental Setups on Validation Set

Key empirical observations include:

2.1. Deficiencies of Vanilla Cross-Entropy

Baseline U-Net achieved an acceptable Disc Dice of 0.8276 but performed poorly on the Optic Cup (0.3047). Standard cross-entropy was dominated by background pixels, limiting sensitivity to subtle cup margins [6].

2.2. Impact of Class Weighting

Doubling the penalty on Cup pixels (Experiment 2) increased Cup Dice from 0.3047 to 0.4149 (+11.02% absolute gain), confirming that class weighting directs gradient updates toward smaller structures [6].

2.3. Superiority of Compound Loss

Experiment 3 produced the highest scores across all metrics: Disc Dice (0.8787), Cup Dice (0.4667), and Mean Dice (0.6727) [6]. While weighted cross-entropy provided smooth pixel updates, Dice Loss directly optimized spatial overlap, improving boundary alignment [8].

2.4. Persistent Cup Segmentation Challenge

Across all models, Cup Dice remained substantially lower than Disc Dice [6]. This reflects clinical reality: the outer disc margin is clearly demarcated by a scleral ring, whereas the cup is defined by subtle color shifts and blood vessel bending (kinking), making it harder to segment accurately from 2D monocular images [2].

3. Model Selection Decision

Based on validation performance, the Weighted CrossEntropy + Dice Loss (Checkpoint: Epoch 14) model was selected as the final candidate [6]. Model selection was conducted entirely on validation metrics without referencing the test cohort, preserving the test partition as an unbiased benchmark [3].

4. Independent Holdout Test Set Evaluation

The selected model was evaluated on the 146 unseen test samples. The final quantitative test scores are detailed in Table 3 [6]:

Metric Evaluation Index	Test Value (N = 146)	Validation Benchmark (N = 68)	Relative Stability
Optic Disc Dice	0.8905	0.8787	+0.0118 (Slight gain)

Optic Cup Dice	0.4625	0.4667	-0.0042 (Negligible delta)
Mean Dice	0.6765	0.6727	+0.0038 (Stable)

Table 4: Performance Metrics of the Final Selected Model on Unseen Test Samples

The test outcomes confirm that the segmentation pipeline generalizes consistently to unseen fundus images. The Disc Dice score (0.8905) indicates reliable localization of the outer optic nerve head.

However, the lower Cup Dice (0.4625) highlights that cup boundary delineation remains a technical bottleneck for automated CDR estimation [6].

5. Qualitative Error Analysis of Predicted Segmentations

Visual inspection of predicted masks against ground-truth labels revealed several characteristic morphological patterns [6]:

- Optic Disc Boundaries: Disc perimeters were generally delineated cleanly and followed true anatomical margins across various retinal pigmentation levels.
- Peripapillary Atrophy Interference: In eyes presenting peripapillary atrophy (PPA) or scleral crescents, the network occasionally misclassified hypopigmented zones as disc tissue, slightly over-expanding the perimeter.
- Cup Fragmentation: The network sometimes produced segmented cups broken into disconnected pixel clusters rather than a single convex shape.
- Boundary Instability: For pale fundus backgrounds with weak vascular contrast, the model struggled to separate physiological cupping from true glaucomatous excavation, resulting in ragged borders [2].

V. REPOSITORY STRUCTURE & EXPERIMENTAL ARTIFACTS

All training weights, metric logs, and generated visual masks are archived in Google Drive at /content/drive/MyDrive/ET4248E_Glaucoma/UNet_Results. Key project artifacts include [6]:

- best_unet.pt: Checkpoint weights for Baseline U-Net (Epoch 29).
- best_unet_weighted.pt: Checkpoint weights for Weighted Cross-Entropy (Epoch 7).

- best_unet_combined.pt: Checkpoint weights for the selected Weighted CE + Dice Loss model (Epoch 14).
- validation_model_comparison.csv: Validation evaluation logs across all configurations.
- final_test_summary.csv: Aggregated test performance metrics across all 146 samples.
- test_per_image_dice.csv: Per-image Disc Dice and Cup Dice metrics for every test sample.
- Training Logs & Plots: Automated execution logs and loss curve visual plots.

VI. PROJECT MILESTONES, CURRENT LIMITATIONS, AND FUTURE WORK

1. Summary of Completed Milestones

The project has completed the foundational stages required for automated optic nerve head segmentation [6]:

- Milestone 1: Verified and paired 485 RIM-ONE DL samples, maintaining label integrity by retaining outlier masks under the Cup Priority Rule.
- Milestone 2: Implemented an end-to-end preprocessing, spatial scaling (256 x 256), and 3-class target generation pipeline.
- Milestone 3: Developed and trained a 482,067-parameter U-Net model in PyTorch.
- Milestone 4: Benchmarked three loss configurations, identifying the hybrid Weighted CE + Dice Loss as the top-performing setup.
- Milestone 5: Evaluated generalization performance on 146 independent test samples and performed a qualitative error audit.

2. Incomplete Tasks and Methodological Limitations

While the segmentation phase is complete, several components required for an automated screening system remain in progress [6]:

- Automated CDR Computation: Direct vertical diameter extraction from predicted masks has not yet been implemented.

- CDR Error Quantification: Statistical error metrics (Mean Absolute Error - MAE, Root Mean Squared Error - RMSE, and correlation coefficients) against reference ground truth have not yet been computed [2].
- End-to-End Glaucoma Classification: Downstream classification (e.g., Logistic Regression or SVM) using model-predicted CDRs has not yet been integrated and tested [3].
- Patient-Level Data Isolation: Due to metadata constraints in RIM-ONE DL, splits were organized at the image level rather than the patient level [5].

3. Action Plan and Future Development

The subsequent phase of this project will focus on the following steps:

- Step 1 (Vertical Diameter Boundary Extraction): Implement bounding-box and contour-scanning algorithms to extract vertical diameters (V_{cup} and V_{disc}) from reconstructed binary masks to calculate:

$$\text{vCDR}_{\text{pred}} = \frac{V_{\text{cup}}}{V_{\text{disc}}}$$

- Step 2 (Morphological Post-Processing): Apply convex hull fitting, morphological opening/closing, and connected component analysis to resolve cup fragmentation and eliminate isolated noise clusters.
- Step 3 (Statistical Error Benchmarking): Calculate MAE and RMSE against expert reference measurements across the 146 test samples.
- Step 4 (Clinical Triage Classifier): Train a classification model using predicted vCDR to evaluate sensitivity, specificity, and ROC-AUC for binary glaucoma screening.

VII. CONCLUSION

This project designed and validated an automated U-Net segmentation pipeline for the Optic Disc and Optic Cup using the RIM-ONE DL dataset. Experimental evaluation demonstrated that combining Weighted Cross-Entropy with Soft Dice Loss achieved the best validation metrics: Disc Dice = 0.8787, Cup Dice = 0.4667, and Mean Dice = 0.6727. When evaluated on 146 unseen test images, the selected model maintained consistent performance with a Disc Dice of 0.8905 and a Cup Dice of 0.4625 [6].

These results show that while optic disc boundary delineation is handled well by the network, segmenting the optic cup remains a technical challenge that impacts automated CDR measurement. Future work will focus on morphological post-processing, computing automated vCDR error metrics, and evaluating the complete screening classification workflow [3].

REFERENCES

[1] World Glaucoma Association, "World Glaucoma Week: For the media," 2025. Available: https://www.worldglaucomaweek.org/for-the-media/ (accessed 20 Oct. 2026).

[2] European Glaucoma Society, "Terminology and Guidelines for Glaucoma, 4th Edition—Part 1," British Journal of Ophthalmology, vol. 101, no. 4, pp. 1–195, 2017. https://doi.org/10.1136/bjophthalmol-2016-egsguideline.439 .

[3] A. S. Panwar, P. Saxena, and V. Sharma, "Clinical Perspectives on the Use of Computer Vision in Glaucoma Screening," Frontiers in Medicine, vol. 11, art. 10972254, 2024. https://doi.org/10.3389/fmed.2024.10972254 .

[4] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in Medical Image Computing and Computer-Assisted Intervention—MICCAI 2015, Lecture Notes in Computer Science, vol. 9351, pp. 234–241, 2015. https://doi.org/10.1007/978-3-319-24574-4_28 .

[5] F. Fumero et al., "RIM-ONE DL: A Unified Retinal Image Database for Assessing Glaucoma Using Deep Learning," Image Analysis & Stereology, vol. 39, no. 3, pp. 159–167, 2020. https://doi.org/10.5566/ias.2346 .

[6] Group 12, "Glaucoma Detection Using the Cup-to-Disc Ratio in Fundus Images," Presentation and associated PyTorch experimental notebooks, Project ET4248E, Hanoi University of Science and Technology, 2026.

[7] J. B. Jonas et al., "Optic disc morphometry for glaucoma diagnosis: Glaucomatous optic neuropathy and cup-to-disc ratio," Survey of Ophthalmology, vol. 43, no. 4, pp. 293–320, 1999. https://doi.org/10.1016/S0039-6257(98)00049-6 .

[8] F. Milletari, N. Navab, and S. A. Ahmadi, "V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation," in 2016 Fourth International Conference on 3D Vision (3DV), pp. 565–571, 2016. https://doi.org/10.1109/3DV.2016.79 .

[9] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in 3rd International Conference on Learning Representations (ICLR), 2015. https://arxiv.org/abs/1412.6980 .